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Predict neuromuscular performance in human epidural electrical stimulation: phase 1 trial interim results

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Abstract

Background Epidural electrical stimulation (EES) has emerged as a promising therapy for restoring motor function in patients with paralysis. A primary challenge in this therapy lies in identifying feasible stimulation parameters in huge selection space for different movements, given the limited understanding of the precise alignment between stimulation and corresponding neuromuscular performance. We aimed to develop a computational framework that predicts neuromuscular performance under EES and thereby reduces the need for extensive in-clinic parameter searches.

Methods We implanted purpose-designed 32-contact epidural interfaces in two individuals with motor-complete spinal cord injury and reconstructed personalized spinal anatomies from medical imaging. Finite element simulations and axonal recruitment modeling were integrated with machine learning to establish a predictive mapping between stimulation parameters and muscle responses. A dimensionality-reduction Bayesian optimization algorithm was subsequently applied to identify compact sets of stimulation parameters targeting specific motor objectives, and selected configurations were validated through clinical testing. This study is an interim report of an ongoing registered clinical trial (Closed-loop Functional Spinal Cord Stimulation in Patients with Spinal Cord Injury, ClinicalTrials.gov Identifier: NCT04969042), sponsored by Beijing PINS Medical Co., Ltd.

Results Here we present purpose-designed 32-contact epidural neural interfaces that enabled two individuals with spinal cord injury to regain lower-limb motor function. The hybrid predictive model demonstrated strong quantitative agreement with experimentally measured muscle responses (mean squared error = 0.0096). Algorithm-guided parameter recommendations comprising 180 configurations outperformed both the historical dataset (1,602 configurations) and conventional bipolar settings across four functional objectives. Clinical validation further confirmed that the recorded muscle activations were in close agreement with model predictions.

Conclusions The AI-aided computational framework can serve as a reliable and feasible agent for evaluating and recommending effective EES parameters. By bridging anatomical modeling with functional outcomes, this approach offers a practical pathway toward optimizing neuromodulation therapies and advancing the development of personalized treatment strategies for individuals with spinal cord injury.

Plain language summary

People with spinal cord injury often lose the ability to move their legs because the connection between the brain and spinal cord is damaged. Over recent decades, epidural electrical stimulation—delivering pulses of electricity to the spinal cord—has been shown to help restore leg movement. However, finding the proper stimulation settings for each person is usually time-consuming and depends on trial and error. In this study, we designed and manufactured a 32-contact spinal implant and implanted it in two people with spinal cord injury. We also developed a personalized hybrid model that combines computer simulations and neural-network predictions to estimate how different stimulation settings affect muscle activity. An optimization algorithm then recommends parameter sets for specific movements, and these recommendations were tested in the clinic. By automating and personalizing parameter selection, this approach can reduce the testing burden on patients and clinicians and help enable more tailored neuromodulation treatments.

Epidural electrical stimulation (EES) of the spinal cord has been shown to be a promising therapy to restore motor function for patients suffering from motor deficits such as spinal cord injury (SCI)[1–11], stroke[12], and other diseases[13,14]. To achieve natural movements, motor activation of targeted joints and muscles are necessary[8,10]. It relies on the recruitment of specific afferent fibers in dorsal roots, simultaneously with the silence of non-targeted nerve fibers to avoid unexpected activation of lower limb muscles[8,10,15,16]. The synergistic activations in EES are realized by a spinal

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interface, where pre-programmed electrical pulses are delivered sequentially via an implanted electrode array[8]. Neuromodulation is evolving toward neural interfaces with more controllable parameters and greater flexibility in stimulation configurations[17]. Utilizing neural interfaces with increased contacts[18] or optimized contact distributions[10] enhances the spatial selectivity of stimulation, potentially enabling a wider variety of neural activation patterns.

However, the increased tuning freedom introduces complexity challenges in determining feasible stimulation parameters for different motions[8,10,19]. Due to the limited understanding of how specific spinal cord segments or nerve roots contribute to the activation of individual muscles, empirical knowledge and manual adjustments still play a dominant role in the parameter-tuning process[7,20,21]. Assessing nerve root activation can be challenging, especially with unexplored stimulation configurations. Furthermore, the intricate spatial mapping between nerve roots and muscles also introduces uncertainty, as a single nerve root often modulates multiple muscles, while a muscle may also be innervated by several nerve roots[22–24]. These multilevel interrelationships between stimulation and effects lead to extensive clinical tests to find feasible stimulation configurations for various motions. Neural interfaces are evolving in complexity to enable more selective modulation of specific target areas. Electrode arrays with 32 contacts have more than 10^{15} polarity combinations for clinicians to set[25]. Besides, amplitudes, frequencies, and pulse widths of the stimulation also need to be tuned and selected to improve rehabilitative performance[8,10]. In our clinical practice, achieving selective activation of targeted neural structures within such a huge space always brings a heavy burden to both clinicians and patients. Significant individual differences[23,24,26] and discomfort caused by overly high stimulation amplitudes also complicate the parameter-tuning process. Multidisciplinary collaborations have enabled rapid functional recovery—such as standing and walking within one single day post-implantation—through highly refined and individualized stimulation protocols. Preoperative imaging as well as personalized reconstruction and simulation of spinal cords were used to help determine implantation locations of the neural interfaces and give insights to the selection of active contacts, thereby providing customized strategies for EES[8,10,11]. These developments have significantly narrowed the practical stimulation parameter space and reduced reliance on trial-and-error methods. However, establishing a more reliable and quantitative mapping between multidimensional stimulation parameters and muscle responses can help to further improve the efficiency of parameter-tuning process.

Here, with the goal of bridging the gap between spinal anatomy and functional innervation in EES, we established the ability to accurately predict neuromuscular performance induced by multidimensional EES parameters using an integrated computational framework. We began by designing and implanting novel 32-contact neural interfaces—customized for motor facilitation rather than pain management—in two individuals with complete motor paralysis[27]. To estimate neural responses evoked by various EES parameters, we reconstructed personalized, anatomy-precise spinal cord models and conducted multi-scale hybrid simulations. We then developed a rapid stimulation-acquisition system to efficiently collect electromyography (EMG) responses corresponding to various EES parameters while assuring feasibility, which served as training data for accurate machine learning. Using these data, our hybrid models could provide reliable quantitative predictions of EMG responses of multiple lower limb muscles. Furthermore, we employed a dimension-reduction Bayesian optimization algorithm to recommend efficient parameters for various targeted movements. By transferring the intensive empirical parameter-tuning tests to computational models, the AI-aided framework could streamline clinical testing processes and reduce potential discomfort issues when exploring unknown parameters, thus alleviating the burden on clinicians and patients. Together, this framework provides a practical and scalable solution for personalizing EES therapy and offers insights applicable to broader neuromodulation interventions.

Methods

Study design

The objective of this study was to optimize the neural interface design for EES based on modeling and simulation as well as to predict the stimulation effects of different stimulation parameters by personalized modeling of EES. All experiments were carried out as part of the ongoing clinical feasibility study "Closed-loop Functional Spinal Cord Stimulation in Patients with Spinal Cord Injury" (https://clinicaltrials.gov/study/NCT04969042), which aimed at evaluating the effectiveness of functional spinal cord stimulation in facilitating ambulatory rehabilitation post-SCI. This study is registered with ClinicalTrials.gov with the NCT number NCT04969042. The date of registration was July 6th, 2021.This study was approved by Ethics Committee of Beijing Tsinghua Changgung Hospital (Approval No.21161-2-02). The present study represents an interim report from this ongoing clinical trial. Two male participants were enrolled in this study. Participant P1 was a 26-year-old male with motor and sensory incomplete SCI, classified as AIS-B at the T5 level, resulting from a car accident 8 months prior to his enrollment. He was unable to ambulate even with the assistance of a walker or braces, resulting in a Walking Index for Spinal Cord Injury (WISCI) score of 0. Participant P2 was a 31-year-old male with motor and sensory incomplete SCI, classified as AIS-B at the T8 level, resulting from a fall from the second floor occurring 26 months prior to his enrollment. He presented with a lower extremity motor score of 0. Although unable to walk independently, he could ambulate using a walking frame and braces without therapist assistance, achieving a WISCI score of 9. Written informed consent was obtained from all the patients to participate in the study. All procedures were performed in accordance with the Declaration of Helsinki and relevant institutional guidelines. Both participants underwent a comprehensive preoperative assessment, which included medical history, neurological examination, and high-resolution 3 T MRI. After the assessment, both participants underwent implantation of the optimized 32-contact electrode array in the spinal epidural space, along with two implanted pulse generators (IPG, investigating model G122RS, PINS). All surgical and experimental procedures were performed at the Beijing Tsinghua Changgung Hospital (BTCH). A postsurgical CT was obtained to confirm the implanted position of the electrode array. Personalized models of the participants' spinal cords were developed based on MRI and CT. To date, no adverse events have occurred, and the devices have not been explanted. To collect stimulation effects evoked by different stimulation parameters, we developed an automated stimulation-acquisition system that could deliver electrical pulses and record corresponding multi-channel EMG feasibly and effectively. We conducted simulations and trained neural networks to predict the stimulation effects of different stimulation parameters. We designed four objectives calculated from EMG and employed a dimension-reduction Bayesian optimization algorithm to search for well-performing parameters based on the model's prediction. The recommended parameters were filtered by clinicians and delivered to participants to validate the model's capability. Written informed consent was obtained from the participants for the publication of their anonymized information in this article.

Geometry of the average model of the human spinal cord and the electrode array

The geometry of the average model of the human spinal cord was derived from anatomical statistics[28–33]. Considering the lower limb muscles are mainly innervated by motor neuron pools and nerve roots in the lumbar enlargement, we modeled low thoracic (T12), lumbar (L1-L5), and sacral (S1 and S2) segments. The lengths of segments were modeled according to reported data[28,29,34] (Table S4). The gray and white matter contours of the average spinal cord model for different segments were derived from anatomical atlases[30]. Specifically, we extracted representative cross-sectional contours for the cervical, thoracic, lumbar, and sacral segments and adjusted their transverse and anteroposterior diameters based on statistical data[30]. These contours were then arranged along the axial direction according to the segmental lengths using SolidWorks, and 3D models of the white and gray matter were reconstructed via the "loft" operation. The cerebral spinal

fluid (CSF) was modeled as an elliptical column surrounding the spinal cord, with its transverse and sagittal diameters determined by the corresponding dimensions of the vertebral canal[35]. The dura mater was modeled as a 0.5 mm thick layer surrounding the outer surface of the CSF. The approximate trajectories of the nerve roots were determined based on their anatomical entry points into the spinal cord and their relative positions to the intervertebral foramina. In the entrance region where each nerve root enters the spinal cord, we constructed bundled structures[10]: each root splits into ten rootlets that enter the spinal cord at an angle of approximately 60 degrees[36] (Fig. 2a). The spinal root entry points were determined from axial MRI slices as the intersections between the spinal cord and the most rostral portion of each spinal root. Between this entry point and the entry point of the adjacent nerve root, we uniformly distributed nine additional points to represent the individual rootlet entry sites, resulting in a total of ten rootlets per root in the model. The number of modeled rootlets exceeds the anatomical averages[37], which could provide finer spatial sampling and smoother recruitment–intensity curves without affecting the physiological interpretation of recruitment proportions. The rootlet diameter measures 0.5 mm in segments T12 to L2 and 0.3 mm in segments L3 to S2[38]. The total length of the modeled structure is 82.25 mm (Table S3), with the assumption that the average total length of the whole spinal cord is 42 cm[28,29,34] and the T12-S2 segments account for 19.7%[33].

Tissues directly affecting the current path between the electrode and the spinal cord, such as the dura mater, were included in the model, whereas tissues with negligible influence on the local current distribution (e.g., bone) were omitted[8,15,39–42].

The paddle lead was modeled (containing metal contacts and the rubber substrate) according to its actual size and placed at the midline outside the dura mater. The contacts were modeled as 3.5×1.0 mm metal sheets, tightly attached to the dura mater. We placed the spinal cord and the paddle lead in a saline cylinder (0.2 m in diameter and 0.3 m in height) to simulate the in vivo environment[15,43–45].

FEM calculation

We used FEM to derive electric fields induced by EES. During the simulations for neural interface optimization, the size of electrode arrays varies within a certain range (Fig. 2), while the geometric center of the electrode arrays remained fixed (as shown in Fig. S1a). The specific conductivity values of the modeled tissues are shown in Table S1[46]. The spinal nerve roots were modeled with the same conductivity values as white matter. To represent anisotropic conductivity, we used a diagonal conductivity tensor matrix, assigning principal conductivity along the rostrocaudal direction.

A current source was set on the surface of the cathodes while the anodes were set to be grounded. Dirichlet boundary conditions were set at the outermost boundaries of the model to simulate the case that the shell of IPG in the distance was set to be grounded:

$$V_e(\delta\Omega) = 0 \tag{1}$$

where δΩ is the outermost surface (saline) of the model. Electrical potential at each position was derived by solving the quasi-static approximation of the Maxwell Equations:

$$\triangle \cdot \sigma \triangle V_e = 0 \tag{2}$$

A nonuniform, second-order tetrahedral mesh was generated for the discretization of the model. The constructed mesh consists of approximately 2,000,000 elements. The simulation was performed using COMSOL 5.5 with a 3.80 GHz Intel i7-10700K CPU with 32 GB RAM.

Evaluation of stimulation selectivity

To achieve more selective stimulation, we hope the electric field induced by EES can be more focused. We defined a selectivity index to evaluate the spatial selectivity of electric fields with reference to previous studies[15,45]. Considering that the spinal innervation region corresponding to a given muscle is typically clustered in distribution, we spatially divided the spinal cord into segments representing the approximate anatomical regions corresponding to different muscles based on literature describing the segmental innervation patterns of lower limb muscles[8,22,23]. We calculated the activation ratio of the repartitioned region based on the activating function, using this as a means to evaluate the spatial distribution of the electric field. Six groups of muscles were considered: iliopsoas (IL), vastus lateralis and rectus femoris (VL&RF), tibialis anterior (TA), biceps femoris muscle and gluteus maximus (BF&GLU), semitendinosus (ST), and gastrocnemius (GM) (Fig. S2). The selectivity index for the *i*th muscle (group) was defined as follows:

$$\mathrm{SI}_i = \mu_i - \frac{1}{m_{neighbor} - 1} \sum_{j \neq i}^{m_{neighbor}} \mu_j \tag{3}$$

where $m_{neighbor}$ represents the number of muscles whose motor neuron pools are adjacent to the *i*th muscle's. The selectivity index ranges from −1 to 1, where −1 represents the maximum activation of all undesired muscles with a complete absence of activation of the targeted muscle, 0 indicates that all muscles are activated at the same level, and 1 means the targeted muscle is activated at the greatest extent while no undesired muscles are activated. And μ_i is the normalized activation of the *i*th muscle and is defined as follows:

$$\mu_i = \frac{\iiint_{\Omega_i} f(x,y,z)\mathrm{d}x\mathrm{d}y\mathrm{d}z}{\iiint_{\Omega_i} 1\mathrm{d}x\mathrm{d}y\mathrm{d}z} \tag{4}$$

$$f(x,y,z) = f(x) = \begin{cases} 1, \text{if} AF(x,y,z) < AF_{threshold} \\ 0, \text{if} AF(x,y,z) \geq AF_{threshold} \end{cases} \tag{5}$$

Ω_i is the virtual segmental volume of the *i*th muscle (group) in the spinal cord. *AF* is the activating function, defined as the second spatial derivative of extracellular voltage along an axon[47]. The average selectivity index of an electrode array is defined as the average of maximum muscle selectivity indexes of all regions which can be achieved with unipolar stimulation under various stimulation amplitudes:

$$\mathrm{SI}_\mathrm{A} = \frac{1}{m} \sum_{i}^{m_{total}} \max(\mathrm{SI}_i) \tag{6}$$

The average selectivity index SI_A reflects the flexibility of stimulation delivered by the neural interface.

Modeling nerve fibers in white matter and nerve roots

Computational models of myelinated fibers within the white matter, dorsal roots, and ventral roots were developed to simulate EES-induced neural responses. Four fiber populations with diameters of 16.0 μm, 10.0 μm, 7.3 μm, and 5.7 μm were implemented using the MRG model to represent a physiologically relevant range of myelinated axon sizes.[43,48,49]. Model files were obtained from Model DB with accession number 3810[50]. For each nerve root, 10 trajectories of fibers were generated whose starting points were distributed uniformly in the entrance region across the cord segment. We modified the Lloyd relaxation method[51] to obtain the distribution of fibers throughout the white matter. We selected several cross-sections of the white matter uniformly along the axial direction. For the most rostral section, we used Lloyd relaxation to uniformly redistribute the points in the white matter. Then the redistributed points were re-centralized and used as the initial seeds to employ Lloyd relaxation algorithm in the other cross sections. There would be a portion of initial seeds outside the white matter area in new sections. We deleted these points and redistributed the remaining points in all sections. The points with the same index were interpolated to generate trajectories of nerve fibers. We adjusted the number used in the most rostral section to generate the desired number of

trajectories. 200 trajectories of nerve fibers in white matter were generated. Electric potentials along each trajectory calculated using FEM were recorded and applied to axon models to estimate the neural responses to EES. For each trajectory, we simulated activation responses for axons of four representative diameters based on the local extracellular potential along the path, with the purpose of assessing diameter-dependent activation patterns. We used NEURON 7.8 solver to carry out simulation of neural activities[52]. A fiber was considered recruited if an action potential traveled along its whole length.

Neural interface optimization

We conducted a comparative analysis of electrode arrays with different geometrical designs using the selectivity index defined by the activating function (AF) as the optimization criterion. The designed electrode array consisted of 32 contacts arranged in a left–right-symmetric, staggered configuration. To evaluate the impact of geometry, we systematically varied the lateral width and rostrocaudal length of the neural interfaces and assessed their selectivity index through simulations. All neural interfaces were positioned at the spinal midline outside the dura mater and centered at the same rostrocaudal level for consistency.

For each electrode configuration, we simulated unipolar stimulation by iterating all active contacts individually and scaling the stimulation amplitude. The selectivity index curves were computed for increasing amplitudes (Fig. S1b), assuming an AF threshold (e.g., 1) to define the activation boundary. Because the potential field generated by a unit current can be linearly scaled, this approach allowed us to estimate the evolution of selectivity across stimulation strengths[53]. For each muscle, the maximum selectivity index across all stimulation amplitudes and all unipolar configurations was identified. The average selectivity index—defined as the mean of these maximal values across all muscles—was then calculated for each electrode design and used as the optimization metric to quantify and compare the overall spatial selectivity of different configurations.

Besides, we employed 2D Bayesian optimization to determine the optimal contact position for unipolar stimulation (Fig. S3). The findings indicate that contact points farther away from the midline can effectively reduce the activation of the contralateral area, aligning with the conclusions drawn from the lateral width analysis. Also, we simulated neural responses of MRG axons in white matter and dorsal roots to explore the influence of the lateral width of the neural interface under unipolar configurations. Recruitment ratios and activation thresholds of different neural structures were calculated under unipolar stimulation using lateral electrode contacts at varying stimulation amplitudes. The stimulation amplitudes were gradually increased with a step size of 0.1 mA, and the activation threshold was defined as the stimulation amplitude at which the first fiber within a given structure was recruited. The Δthreshold represents the difference in activation thresholds between the ipsilateral dorsal root (on-target structure) and either the contralateral root or the dorsal column (off-target structures) under the same stimulation configuration. Figure 2b demonstrates Δthreshold values calculated for 10.0 μm fibers in the L2 dorsal root under single 210 μs monophasic pulses.

A comparison was made between a 32-contact electrode array and two 16-contact arrays, one of which follows the commonly used 5-6-5 electrode array pattern (Fig. 2d). The sizes of the electrode arrays were maintained roughly consistent. To comprehensively evaluate their performance, we explored polarity combinations with certain limitations (1 cathode with anodes no more than 3, or 2 cathodes with anodes no more than 2). The selected configurations are representative, as an excessive number of contacts could lead to diffused stimulation and poor selectivity index performance. Simulation results demonstrate a significant advantage of the 32-contact electrode array, with an average maximum selectivity index (SI_A) of 0.34 compared to around 0.28 for the 16-contact arrays.

The arrangement of contacts was based on the physiological characteristics of the human spinal cord (Fig. 2e). Previous studies indicate that sacral segments are typically shorter than lumbar segments, and nerve roots are more densely distributed in sacral regions[10,32,33]. Therefore, we distributed contacts unevenly along the rostro-caudal direction. The longitudinal spacing between contacts increases linearly from sacral regions to lumbar regions, aiming to enhance stimulation selectivity in sacral segments.

The results for lateral width, rostrocaudal length and contact number were based on simulations with homogeneous contact spacing. Simulations we conducted indicate that changing the contact distributions to an inhomogeneous distribution has minimal impact on the overall selectivity index (Fig. S5).

Based on preoperative spinal cord reconstruction and simulation results, we designed two electrode arrays for two participants. The electrode array for P1 measures 52.25 mm long and 7.1 mm wide, while P2's array is 66.75 mm long and 10.0 mm wide. The increments of longitudinal spacing for the two arrays are 0 mm and 0.25 mm.

Electrode array fabrication and control

The optimized electrode array was fabricated using the same material and process as a commercial 16-electrode array (Beijing PINS Medical Co., Ltd., model L3253). The electrode contacts were made of 90Pt/10Ir (3.5 × 1.0 mm), and the substrate of the electrode array was made of implantable medical-grade silicone rubber. The mechanical, electrical and biocompatibility properties of the electrode array were validated through extensive testing. Prior to the start of the clinical trial, we commissioned the Beijing Institute for Medical Device Testing of China National Medical Products Administration (NMPA) to conduct testing and inspection of the IPG and neural interfaces. The test report number is W-W-2930-2020 and the test results met the requirements of the ISO 14708−1:2000 standard. The neural interface was connected to and controlled by two 16-channel IPGs (Beijing PINS Medical Co., Ltd., model G122RS), which support wireless charging and signal acquisition functions. All medical devices used were approved by the ethics committee of Beijing Tsinghua Changgung Hospital. The IPG used in this study operates in current-controlled mode. During multi-anode stimulation, all anodes share the same potential and controlled collectively by the current source. In most cases, the maximum stimulation parameters used in the clinical trials were below 4 mA in current and 300 μs in pulse width, resulting in a maximum charge density of less than 0.034 mC/cm². This maximum value remains close to the reported safety limit for PtIr electrodes[54,55].

Personalized modeling of spinal cord

The geometry of the personalized spinal cord models was primarily constructed based on two types of preoperative MRI sequences (Table S5). This anatomically accurate volume conductor model encompasses gray matter, white matter, cerebral spinal fluid (CSF), dura mater, and nerve roots. From axial images acquired at different vertebral levels, we manually extracted the contours of the white matter and CSF. These contours were then used in SolidWorks to perform a lofting operation and generate three-dimensional models of the white matter and CSF. The trajectories of the spinal roots were also manually traced from the axial MRI images. The spinal segment corresponding to each root was identified based on the exit location of the root from the spinal canal. At the entry region of each root into the spinal cord, we constructed a root bundle in which individual rootlets were evenly distributed within the corresponding spinal segment. For each spinal root, 10 rootlets were modeled. The rootlet construction method was consistent with that used in the average model.

The entire ensemble model was placed in a saline conductor to simulate the human body environment. MRG models of nerve fibers were incorporated into the nerve roots and white matter.

For P1, the personalized model includes segments from T11 to S2, with the targeted segment length (L1 to S2) measuring 47.6 mm. For P2, the personalized model encompasses segments from T12 to S2, with the targeted segment length measuring 52.3 mm. The position of the implanted electrode array was determined from postoperative CT scans of the patient.

Resistance adjustment of the model

We adjusted the resistance between contacts in models to ensure the accuracy of electric simulation (Fig. S19). The 32-contact electrode array is controlled by two IPGs, with each IPG connected to 16 contacts on either the left or right side. The resistance between every two contacts controlled by the same IPG was measured by delivering a 1 mA current pulse with a pulse width of 210 μs. The voltage difference between the two contacts was recorded within 30 μs after the rising edge of the pulse, and the resistance was calculated accordingly. We represent these resistances using a 2D matrix, where the value at the nth row and the mth column corresponds to the resistance between contact n and contact m. We recorded the resistance matrices of two participants when the measured resistance had stabilized, approximately two weeks after the surgery. To align with experimental measurements, a virtual resistance was introduced between each contact and the dura mater in models (Fig. 3b, S10, S11).

Two weeks after the implantation surgery, we could calculate the resistance between every two contacts by applying unit currents sequentially and measuring the potential difference. The resistances between each contact and the outer case of the IPG could be calculated in the same way. The resistance between contacts could influence the current distribution (Fig. S19) thus we adjusted the modeled resistance to simulate the realistic scenario. We assumed that the resistance between the contact and the outer case of IPG was composed of two parts: the local resistance between the contact R_i^{IPG} and the tissue and the resistance between the tissue and the outer case of the IPG R^t. Also, we assumed the resistance between two contacts were composed of two local resistances between the contact and the tissue as well as the initial modeling resistance between two contacts $R_{i,j}^{initial}$, which could be calculated from FEM simulation. Further, we assumed that the resistance between the tissue and the outer case of IPG is approximately equivalent for each contact. Under this assumption, the local resistance for each contact could be calculated by:

$$R_i^{local} = R_i^{IPG} - R^t \tag{7}$$

The resistance between two contacts could be calculated by:

$$R_{i,j} = R_i^{local} + R_j^{local} + R_{i,j}^{initial} \tag{8}$$

R^t was adjusted to obtain the local resistance R_i^{local} for each contact to minimize the calculated $R_{i,j}$ from Eq.(8) and the measured resistance between contact i and contact j. Then we added the local resistance R_i^{local} to the corresponding contact in the model by embedding a 0.02 mm thick slice in the contact. The thin sheet has the thickness of 0.02 mm and the conductivity values were calculated from R_i^{local}. Then we recalculated the resistance between every two contacts in the model after the adjustment to obtain the modeled resistance matrix. Despite the assumptions and simplifications used in this process, the obtained modeled resistance matrices show good consistency with the measured resistance matrices (Fig. 3b, S10, 11).

Calculation of activation matrix

For each stimulation configuration, electric fields were calculated with COMSOL 5.5. Electric potentials along fiber trajectories were recorded and interpolated from the FEM solution. For each fiber trajectory, we constructed the stimulation as square pulses and applied the stimulation array to fiber models with 4 diameters (5.7, 7.3, 10.0, 16.0 μm) using the "play" method in NEURON[52]. The simulation was conducted with a time step of 5 μs and the simulation time was set to be 8 ms. We calculated the recruitment ratio for 18 structures in the spinal cord, including 16 dorsal roots (the dorsal T12-S2 roots of both sides, each root contains ten fiber trajectories), as well as the left and right portions of the white matter (each portion contains 90 fiber trajectories). Thus, we obtained an 18×4 matrix, which comprises the recruitment ratios for 18 neural structures and fibers of 4 diameters. We supposed the activation matrix could represent the neural responses and spatial information for each EES configuration and used it as the input of the multi-layer perceptron.

Automated rapid acquisition of EMG responses

To efficiently and reliably deliver stimulation and record corresponding EMG, we developed an automated stimulation-acquisition system. An upper computer connected and controlled the IPG via Bluetooth, sending commands such as "stimulation on" or "stimulation off" at specified time intervals. These intervals were set at 800 ms, indicating the stimulator would stimulate for 800 ms and then rest for 800 ms. The upper computer could also update stimulation parameters while sending "stimulation on" instructions. Stimulation frequency was set at 2 Hz to ensure a complete single pulse delivery within the stimulation-on period (Fig. S12).

The automated stimulation-acquisition process was performed in two stages. In the first stage, bipolar stimulation was used to determine approximate tolerance thresholds. In this phase, a lateral contact served as the cathode and the adjacent contact in the same row served as the anode. The pulse width (PW) was set to 210 μs for P1 and 300 μs for P2, the amplitude was increased from 0 mA in 0.1 mA steps, and the inter-pulse interval was 1.6 s. Stimulation was stopped when either a noticeable movement was produced by the participant or the EMG of a given channel no longer increased with higher stimulation amplitudes. This procedure was repeated across the eight lateral contacts, and the final stimulation amplitudes for each contact were recorded.

During the second stage, a list of stimulation parameters was randomly generated and delivered to the IPG. Parameters included various polarity combinations, pulse widths, and amplitudes, constrained for risk control based on the pre-acquisition results and clinician guidance. For P1, most trials used a fixed PW of 210 μs and amplitudes ≤ 4.0 mA; for P2, PW values were randomly selected from 100 to 500 μs and amplitudes ≤ 3.0 mA. Additionally, parameter sets with fixed polarity combinations and with pulse width varied incrementally from 100 to 1000 μs or amplitude varied incrementally were included to obtain a more comprehensive dataset.

Throughout the procedure, participants remained lying in bed and were asked to relax. Surface EMG was recorded from 12 muscles bilaterally: iliopsoas (IL), rectus femoris (RF), tibialis anterior (TA), biceps femoris (BF), gastrocnemius (GM), and gluteus maximus (GLU), using the Noraxon Ultium System at a sampling rate of 2000 Hz. This automated system enabled rapid acquisition of EMG responses across a wide range of stimulation parameters without manual intervention.

Stimulation with bipolar configurations was also delivered to participants using the automated stimulation-acquisition system (Fig. S13–15). One of lateral contacts served as the cathode, while the neighboring contact in the same row acted as the anode (Fig. 4d). Amplitudes were increased incrementally from 0 until the induced EMG ceased to increase or until the clinician decided to stop. We also set an emergency button that can stop stimulation immediately to further ensure feasibility. No discomfort was reported throughout the procedure. Data collection for P1 occurred in four different days, while all data for P2 were collected within one day.

Data processing

The collected surface EMG signals were band-pass filtered between 20 and 450 Hz using a 4th-order Butterworth filter. We used signals induced by calibration pulses to synchronize stimulation parameters and EMG responses. The peak-to-peak values of EMG were segmented and calculated for corresponding stimulation. We detected and removed the noise caused by movements manually. For P1, signals of left biceps femoris for 77 parameters were lost due to sensor detachment (Table S6). The peak-to-peak values were normalized by the maximum achieved for each muscle during the experimental session.

Correlation analysis between activation matrices and muscle activities

We tried to derive the innervation of different lower limb muscles from the simulation and clinical data. First, we calculated Pearson correlation

coefficients (PCCs) between different muscle activities to obtain muscle synergy information (Fig. S17). Then PCCs were calculated between activation matrices and each channel of EMG signals to obtain the innervation maps for each muscle (Fig. S18). We hypothesized that a higher correlation coefficient indicates a higher possibility that the muscle is modulated by the nerve root. Bipolar stimulation results were used to validate the inferences from calculation (Fig. S13–16).

Neural network training

We used a multi-layer perceptron (MLP) to solve the regression problem (Fig. 3d). The MLP consists of two hidden layers and one output layer, using ReLU and Sigmoid as activation functions, respectively. The input of MLP was activation matrices and output was normalized EMG peak-to-peak values. The data collected during automated acquisition process were randomly divided into the training set (80%) and the test set (20%). Our loss function during the training process consisted of two components: Mean Squared Error (MSE) loss and ordinal loss, defined as:

$$Loss = Loss_{MSE} + Loss_{ordinal} \quad (9)$$

Considering that the data we collected contained a small number of parameters that can induce strong muscle activities, we designed an ordinal loss. For each iteration, 10% of activation matrices were randomly selected and increased along several dimensions. If the increased matrix resulted in a lower output, the ordinal loss would accumulate the reduced value. This approach improved possibilities that higher-amplitude parameters induced stronger muscle activity (Fig. S19b, c). Considering the limited scale of the dataset, we used five-fold cross-validation during the training process, and the predictions of the five models were averaged to get the final prediction.

Model-based parameter optimization and validation

We employed a dimension-reduction optimization algorithm to search for useful parameters based on the prediction of the model[56]. The algorithm aims to identify effective and well-tolerated EES configurations while minimizing the number of required in-clinic tests. It avoids probabilistic risks during the entire search process and can efficiently optimize high-dimensional functions through latent-space embedding and Bayesian optimization.

Each EES configuration was represented by an 18-dimensional vector, where 16 dimensions encoded the polarity combinations, and the remaining two dimensions represented stimulation amplitude and pulse width, respectively. To better capture the spatial distribution of stimulation, each configuration was also transformed into a 52×14-pixel electric field map using a linear diffusion model that simulated field spread from cathodes and anodes over the electrode layout. These field maps, together with pulse widths, were compressed into a 4-dimensional latent space using an auto-encoder trained on 10,000 randomly generated configurations. When optimizing with respect to pulse width, this parameter was concatenated with the encoded field representation. Introduction and codes about the algorithm could be found at HdSafeBO[56].

Within the latent space, a modified Bayesian optimization algorithm was employed to iteratively search for parameter sets that maximized predefined functional objectives (Table S2)[8,10]. Four objectives were defined, corresponding to four lower-limb movements. The optimization started with an initial dataset composed of the historical stimulation-acquisition data and 200 randomly selected configurations. At each iteration, the algorithm proposed a batch of 20 candidate configurations, which were evaluated using the hybrid predictive model. The predicted objective values were then used to update the Gaussian process surrogate, guiding the next iteration of sampling (Fig. S20, S21).

For P1, pulse width was fixed at 210 μs during the optimization process, while polarity combinations and amplitudes were varied to maximize the objective functions. For P2, both pulse width and amplitude were optimized simultaneously. Approximately ten optimization batches were performed for each objective. To ensure feasibility and avoid excessive activation, any configuration with predicted muscle activation exceeding 0.9 in any channel was excluded. From the final recommended batches, a candidate parameter set was selected according to predicted performance and clinical review. This set included 80, 50, 30, and 20 configurations for the four respective objectives (Fig. 6, S22).

The selected parameters were then tested experimentally using the automated stimulation-acquisition system. EMG signals from twelve muscles were recorded, and muscle activities were renormalized if the maximum peak-to-peak amplitude of the candidate set exceeded that of the historical dataset. None of the parameters collected during this follow-up validation were included in the network training dataset, and the predictive model was not re-tuned using newly acquired data.

The influence of postures on the effects of EES

We delivered the same series of stimulation with increasing amplitudes to participants at different postures using an automated acquisition system. The participant was lying in a bed with their torso positioned at various angles relative to the horizontal plane, while their lower limbs remained fixed. The angles were set to 0, 30, 45, and 60 degrees, respectively (Fig. S24). The corresponding muscle activities were recorded and aligned to stimulation parameters.

Statistics and reproducibility

Data computation and analyses were performed using MATLAB R2019b and Python. Pearson correlation coefficients were calculated using the NumPy package (v1.26.4). Mean squared error (MSE) and R^2 values were computed using the scikit-learn package (v1.4.1.post1). For comparisons between recommended parameters and historical clinical parameters, two-tailed Student's t-tests were used to calculate p-values.

Results

Model-based neural interface optimization

First, we developed an anatomically average model of the human spinal cord to calculate electric fields induced by EES (Fig. 1, Fig. 2a). The geometry of the model matches the reported anatomy statistics[28–33]. The computational model encompasses gray matter, white matter, cerebral spinal fluid (CSF), dura mater, and nerve roots of T12-S2 segments, with a total length of 82.25 mm. McIntyre-Richardson-Grill (MRG)[43,48,49] axon models were integrated into the spinal cord model and neural responses of fibers in nerve roots and white matter were simulated (Fig. S1). Previous studies have commonly reported that EES preferentially recruits large-diameter afferent fibers at the dorsal root entry zone[10,15]; therefore, we explicitly modeled the trajectories of the nerve roots within the cerebrospinal fluid. A selectivity index was defined based on the activating function (the second spatial derivative of extracellular voltage along the axial direction[47]) to represent the spatial selectivity of the EES-induced electric fields. Then we conducted the modification of the neural interface for EES using the selectivity index as the criterion based on simulations. We focused on four key aspects: number of contacts, rostrocaudal length, lateral width, and spatial arrangement (Fig. 2). We investigated whether a 32-contact electrode array had better stimulus selectivity than the commonly used 16-contact electrode array. By traversing the contact selection of unipolar configurations, the average of the maximum selectivity indices for six muscle groups was calculated. Compared to the 5-6-5 paddle lead, the 32-contact lead demonstrated an 18.0% improvement in the average selectivity index. Similarly, compared to the 16-contact lead with four rows of contacts, the 32-contact lead exhibited a 28.5% increase in the average selectivity index (Fig. 2d). The rostrocaudal length of the electrode array was defined as the extent from the top of the most rostral contact to the bottom of the most caudal contact. Simulation results indicate that an electrode array slightly longer (with a rostrocaudal length of 74 mm) than the target regions (L1-S2 segments, with a total length of 66 mm) could achieve an optimal average selectivity index of 0.294 among six groups of muscles (Fig. 2c, S4). The lateral width of the electrode array was defined as the extent from the left edge of the leftmost contact to the right edge of the rightmost contact. We demonstrated that a wider

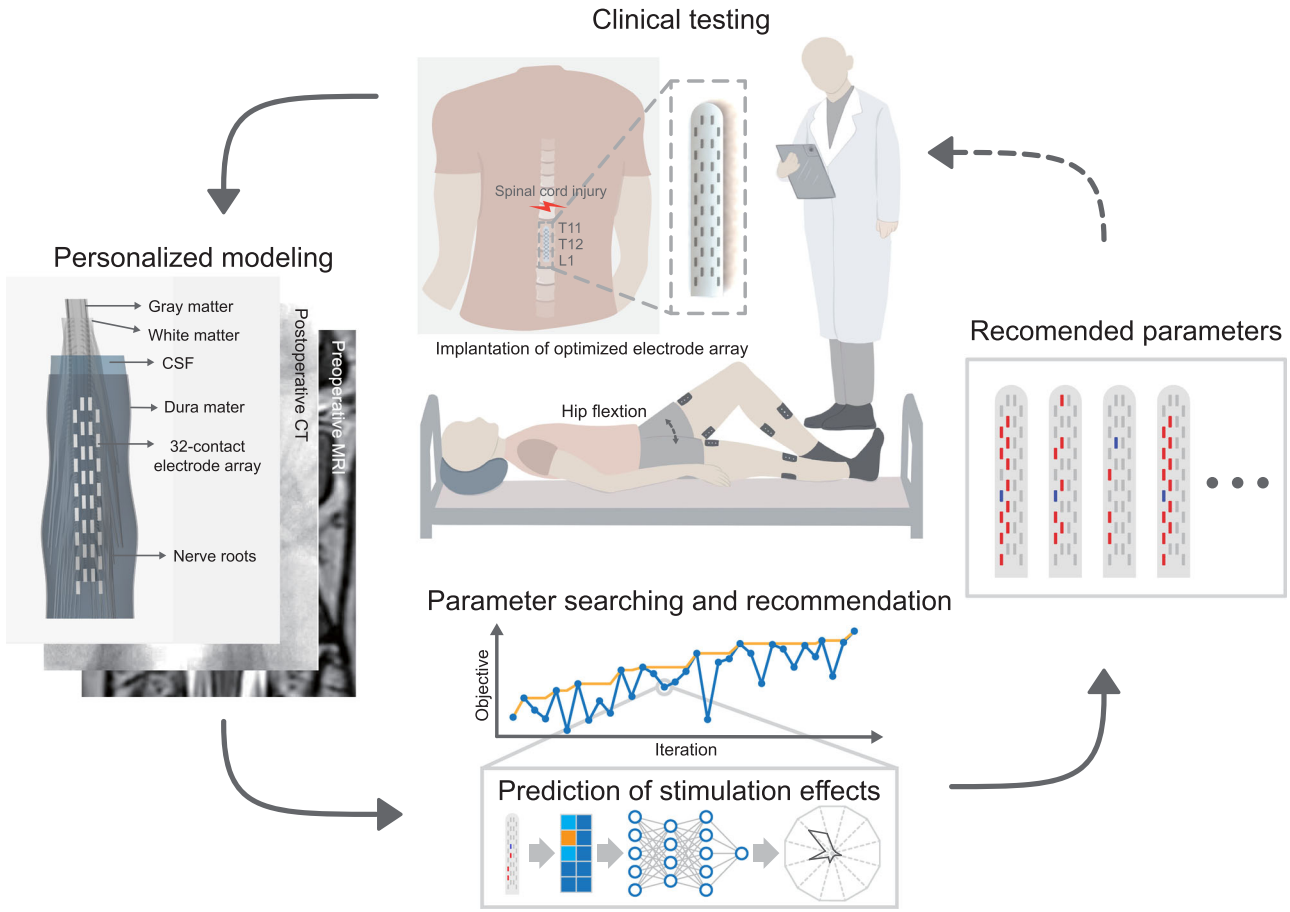


Fig. 1 | AI-aided EES parameter recommendation based on personalized modeling and simulation. After the implantation of the neural interface, personalized EES models were established to predict neuromuscular performance of different parameters in a data-driven manner. A dimension-reduction algorithm is employed to explore and recommend feasible EES parameters for targeted movements. The parameters and corresponding muscle reactions are provided to clinicians as quantitative references to replace intensive onsite tests.

electrode array would achieve a higher average selectivity index (Fig. 2b). When the lateral width was increased from 7.0 mm to 11.0 mm, the average selectivity index increased by 18.5% (from 0.254 to 0.301). Under unipolar stimulation, using the contact further from the midline could improve the activation thresholds of nerve fibers in white matter and contralateral dorsal roots. Finally, we proposed an inhomogeneous longitudinal spacing of contacts according to the physiological characteristics of the spinal cord. The proposed electrode array features higher local contact density in the sacral region than in the lumbar region to match the denser distribution of nerve roots and motor neuron pools in sacral segments (Fig. 2e). Simulation results prove that introducing non-uniform contact spacing does not reduce the average selectivity index of the neural interface (Fig. S5).

We fabricated two types of 32-contact electrode arrays for EES. The smaller one measures 52.25 mm long and 7.1 mm wide, while the larger one is 66.75 mm long and 10.0 mm wide. The increments of longitudinal contact spacing for the two arrays are 0 mm and 0.25 mm (Fig. S6). Based on the MRI-based measurements and assessments of the surgeon, we implanted the smaller interface in P1 and the larger interface in P2. Both participants were suffering motor complete SCI and graded class B by the American Spinal Injury Association (ASIA) Impairment Scale (ASI)[27,57] (Fig. 2f, S7).

Estimation of neural responses in EES with anatomy-precise and resistance-tuned modeling

To quantify the neuromuscular performance elicited by EES, we constructed personalized computational models of the spinal cord for two SCI participants who were implanted with the optimized 32-contact neural interface. The reconstructed models incorporated individual-specific spinal geometries derived from preoperative MRI and CT imaging, including detailed representations of white matter, CSF, and nerve root trajectories (Fig. 3a, Fig. S7). For participant P1, the personalized spinal cord model spanned from T11 to S2, with a targeted segment length (L1–S2) of 47.6 mm. For P2, the model covered T12 to S2, with a targeted segment length of 52.3 mm.

Following model reconstruction, we embedded axon models with four different diameters (16.0, 10.0, 7.3, 5.7 μm) into both the white matter and the nerve roots to represent distinct fiber populations. Using the FEM, we then calculated the electric potentials generated by different EES configurations. After electrode implantation, substantial variability in resistance was observed across contacts. For P1, the maximum and minimum inter-contact resistances were 3009 Ω and 835 Ω, respectively, while for P2, they were 3264 Ω and 976 Ω, differing by over threefold in both cases (Figs. S10, S11). We found that these resistance variations influenced current distribution and consequently altered the electric field patterns generated by EES (Fig. S19). To enhance the simulation accuracy, we incorporated empirically measured inter-contact resistances into the model. These adjustments led to a better match between modeled and measured resistance matrices, with deviations reduced to within 10% (Fig. 3b, Figs. S10, S11).

Using the refined potential fields, we simulated axonal responses and characterized the activation patterns across different stimulation conditions. An 18×4 activation matrix was generated for each configuration, summarizing the recruitment ratios of axons across fiber groups in both the roots and white matter (Fig. 3c). These results support a strong association between EES-induced electric field distributions and the corresponding

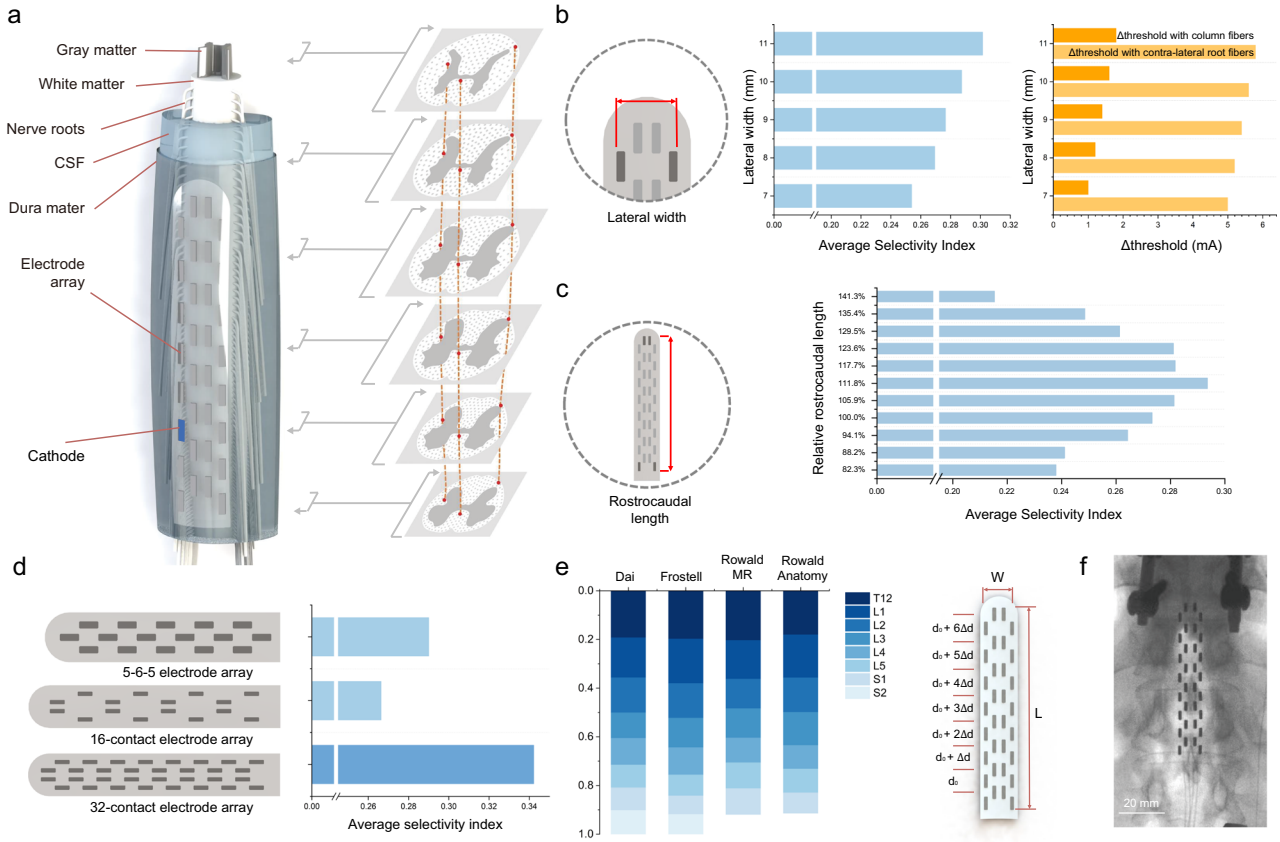


Fig. 2 | Model-based optimization of EES neural interfaces. a average model of the human spinal cord developed based on anatomy statistics. Our optimization focused on four aspects of the neural interface: **b** Lateral width. A wider electrode array could achieve a higher average selectivity index and avoid the activation of contra-lateral nerve roots. Selectivity indices were calculated with the rostrocaudal lengths of the neural interfaces fixed at 52.65 mm. The right panel illustrates the activation threshold difference of 10.0 μm-diameter fibers between the target ipsilateral nerve roots and the dorsal column and the contralateral nerve roots during unipolar stimulation using lateral contacts. **c** Rostro-caudal length. An electrode array slightly longer than the targeted segments has better performance in the simulation. Selectivity indices were calculated with the lateral widths of the neural interfaces fixed at 8 mm. **d** Number of contacts. We compared a 32-contact electrode array with 16-contact electrode arrays. More contacts would improve the flexibility of stimulation. **e** Contact arrangement. As the lumbar segments of the spinal cord are usually longer than the sacral segments, we redistributed the contacts with an uneven design. The longitudinal spacing between contacts increases linearly from sacral region to lumbar region. **f** X-ray imaging of P2 after the implantation surgery.

neural recruitment, providing the basis for downstream EMG prediction and stimulation parameter optimization.

Personalized muscle innervation maps derived from stimulation-response data

To establish a robust mapping between stimulation parameters and muscle activation, we devised a rapid stimulation-acquisition system capable of efficiently and reliably delivering stimulation while recording corresponding muscle activities. Using the implanted 32-contact electrode arrays, we delivered 3624 stimulation configurations to P1 and 4872 to P2, with each configuration applied at a rate of one pulse every 1.6 s. Each configuration included randomized combinations of polarity, amplitude, and pulse width. Surface EMG signals were simultaneously recorded from twelve lower limb muscles bilaterally, including the iliopsoas (IL), rectus femoris (RF), tibialis anterior (TA), biceps femoris (BF), gastrocnemius (GM), and gluteus maximus (GLU) (Fig. S12). Muscle activation was quantified using normalized peak-to-peak EMG amplitudes, and activation matrices were calculated for all tested parameters. The personalized stimulation-response dataset for P1, consisting of EES parameters and corresponding 12-channel EMG recordings, was acquired over four days. For P2, the full dataset was collected within half a day. No discomfort or adverse events were reported during the data acquisition sessions.

Analysis of these data revealed distinct, participant-specific patterns of muscle innervation. By analyzing the correlations between the activation amplitudes of different muscles, we characterized the muscle synergies of each participant under EES. For P1, the tibialis anterior (TA) and gastrocnemius (GM) on both sides tended to be co-activated, with correlation coefficients of 0.84 on the left and 0.97 on the right. Additionally, the iliopsoas (IL) and rectus femoris (RF) also exhibited moderate correlations in activation (0.72 on the left and 0.76 on the right). For P2, a strong correlation was observed between the activations of the left TA and GM (0.97), while the activations of the IL and RF were also highly correlated on both sides (0.88 on the left and 0.83 on the right). By correlating EMG responses with the simulated recruitment ratios of axons in specific nerve roots, we generated innervation maps for each muscle, which revealed the spatial distribution of correlations between muscle activation and neural recruitment along the rostrocaudal axis (Fig. 4, S17, S18). For P2, the inferred innervation maps of the GM were consistent across both sides (Fig. 4c). In contrast, the TA showed a laterality-dependent pattern: the left TA was more strongly associated with S1/S2 root recruitment, whereas the right TA correlated with L4/L5 activation (Fig. 4b).

These inferences were supported by targeted bipolar stimulation (Figs. S13–16). For instance, the use of contacts at the 4th row (4 + 12 −) selectively activated the right TA, while more caudal configurations were required to activate the left TA (Fig. 4d). Additionally, during the stimulation sweep, synchronous activation was observed between the left TA and left GM (Fig. 4a), which was mirrored in their similar innervation maps. These findings suggest functional coordination between the two muscles and validate the use of activation matrices to estimate nerve root-level recruitment underlying EMG responses.

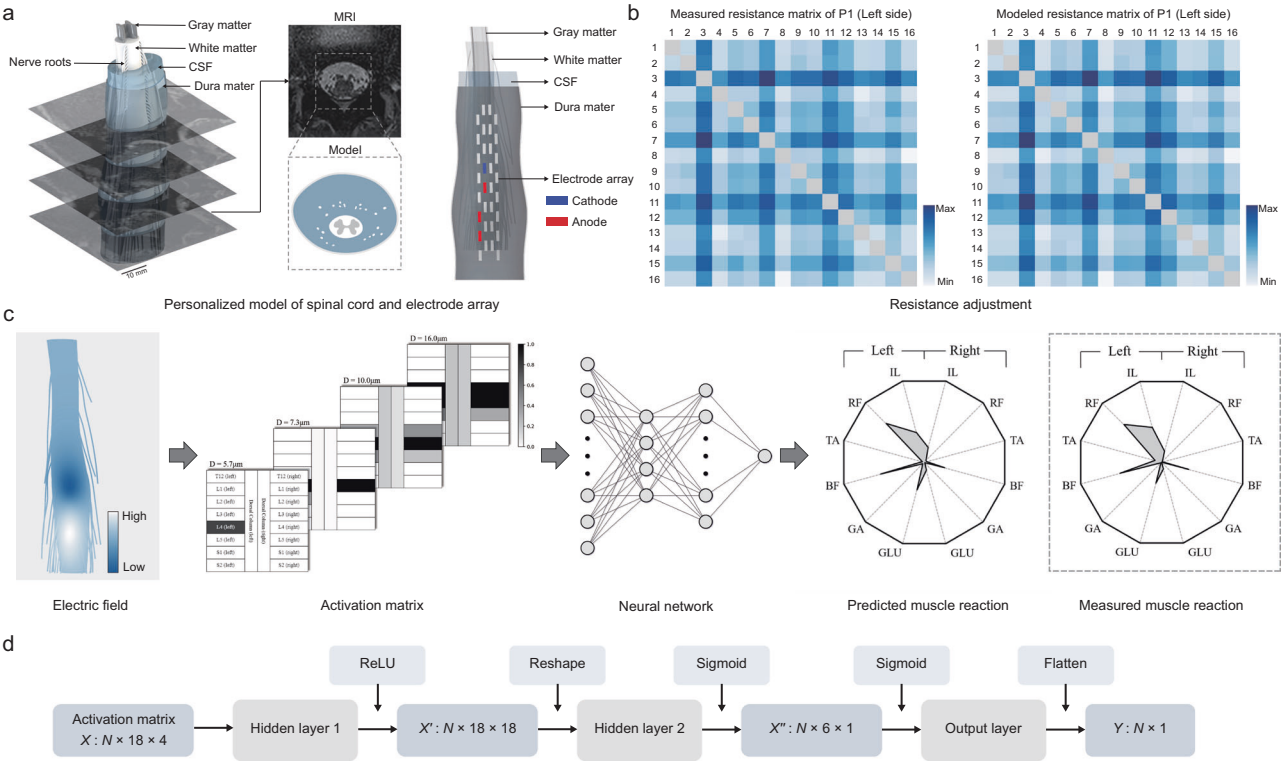


Fig. 3 | Modeling to predict neuromuscular performance induced by various stimulation parameters. a Personalized model of EES. The model developed based on pre-surgery MRI contains gray matter, whiter matter, CSF, dura matter, and nerve roots. The electrode array was located and modeled based on post-surgery CT. **b** Resistance adjustment. The matrix on the left illustrates the measured resistance between contact pairs of the left 16 contacts after the implantation. The matrix on the right displays the resistance matrix in the model after adjustment. **c** Procedures to predict muscle activities induced by EES. Electric fields were calculated using FEM, which was used to simulate neural responses in different structures. Activation matrices were used to represent the activation of fibers in nerve roots and white matter. A neural network was trained to predict the muscle activities. **d** Neural network design. We used a multi-layer perceptron to predict corresponding normalized muscle activities based on the calculated activation matrix. The MLP consists of two hidden layers and an output layer.

Accurate Prediction of muscle performance of multidimensional EES parameters

To enable data-driven prediction of muscle activation to EES, we trained a multilayer perceptron (MLP) for each participant using activation matrices derived from simulations as inputs and the normalized peak-to-peak EMG amplitudes as outputs (Fig. 3d). The trained model demonstrated reliable performance in predicting muscle activation across a wide range of stimulation parameters.

For example, in P1's left iliopsoas, the mean normalized EMG amplitude was 0.271, and the test set yielded a MSE of 0.0149 (Fig. 5a). Similarly, for P2's left rectus femoris, the mean amplitude was 0.247 with a test MSE of 0.0199 (Fig. 5b). Across both participants, Pearson correlation coefficients between predicted and recorded EMG responses exceeded 0.8 for most muscle channels, achieving an average of 0.869 (Fig. 5c, d).

To evaluate classification accuracy, we grouped the stimulation parameters into four classes based on the quartiles of muscle activation amplitude. The resulting confusion matrices indicated that the model accurately differentiated between strongly activating and non-activating parameters, with misclassifications primarily limited to adjacent classes. Detailed results for all muscles are provided in the supplementary materials (Table S6–S9, Figs. S25–48).

We further validated the model performance using independent datasets acquired at later time points (one year for P1, one month for P2). Despite the temporal gap, prediction accuracy remained high, with Pearson correlation coefficients above 0.7 for most channels (Fig. 6, Fig. S23). In addition, the model was tested against a set of bipolar stimulation results that were not included in the training or test datasets. The model successfully predicted activation thresholds and engagement patterns, confirming its generalizability (Figs. S13–16).

Model-based EES parameter recommendation using a dimension-reduction algorithm

Following validation of the predictive model, we employed it as an agent to guide the selection of effective stimulation parameters, aiming to reduce the reliance on extensive in-clinic testing. Considering the complexity of the stimulation parameter space—including polarity combinations, amplitudes, and pulse widths—we developed a dimensionality reduction-based optimization framework to identify candidate settings based on prior data and model predictions (Fig. S20)[56].

We employed the developed algorithm to search for optimal stimulation parameters according to predefined objectives (Table S2) that reflect specific motor functions[8]. The optimization loop iteratively selected candidate configurations, evaluated them via the hybrid model, and updated the Gaussian process surrogate to guide subsequent exploration. Compared to initial random sampling, the recommended parameters consistently achieved higher predicted objective values during optimization (Fig. 6a).

To validate the recommended parameters, configurations which may lead to strong muscle activation were excluded, and a final candidate set was compiled based on predicted performance. These parameters were tested experimentally, nearly one year after the original data collection for P1. Despite this interval, the model's predictions remained consistent with the recorded EMG responses, with an average Pearson correlation coefficient of 0.673 and an average MSE of 0.041 (Fig. 6c, d). The final candidate set, consisting of 180 configurations, outperformed both the historical dataset (containing 1,602 configurations) and bipolar stimulation settings (containing 134 configurations) across all four functional objectives (Fig. 6b). Specifically, the maximum objective values for hip flexion (Objective 1) and ankle flexion (Objective 3) were improved, while the average objective value across all four motor targets increased by 0.156 compared with the historical

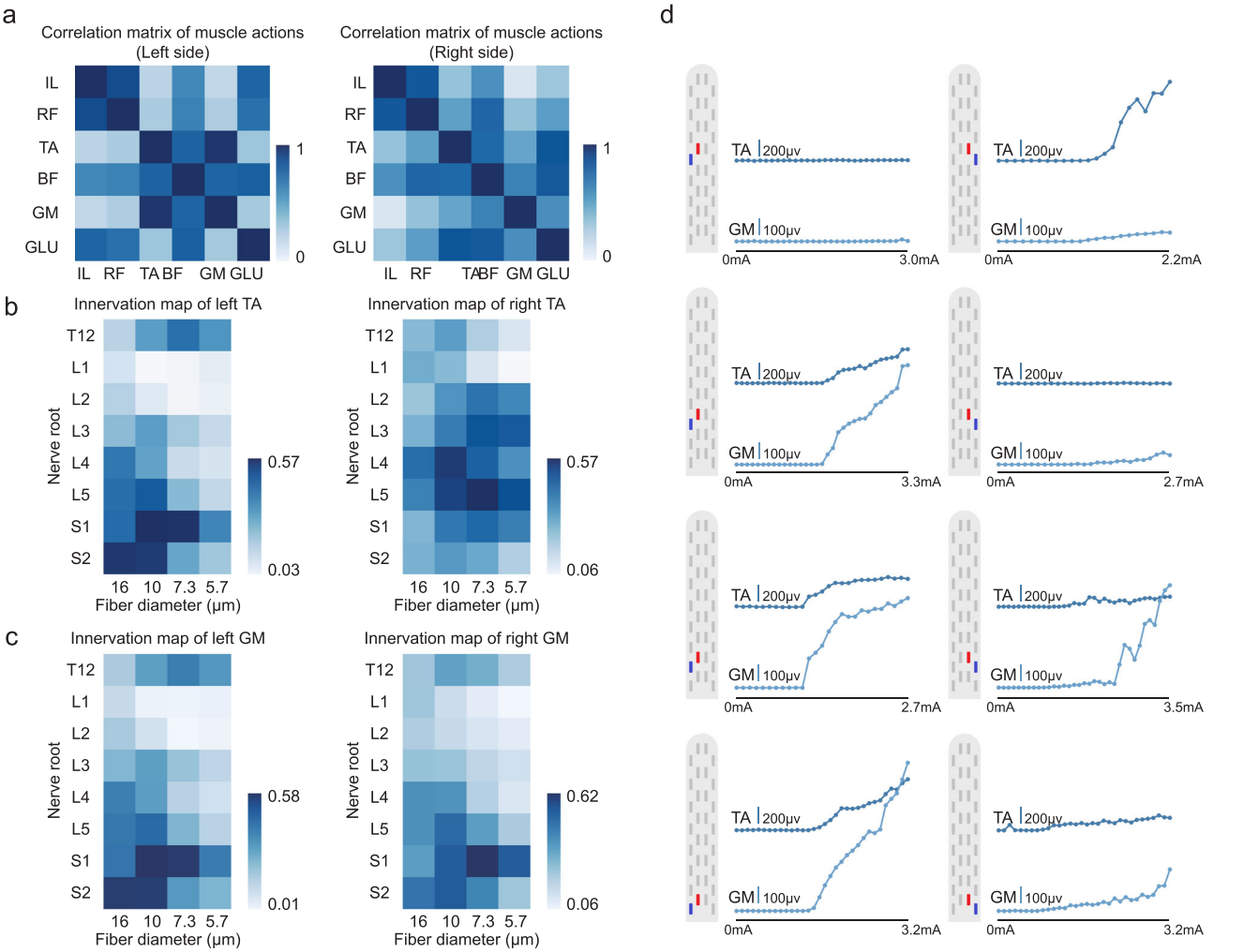


Fig. 4 | Innervation maps between fiber recruitment and muscle activities. **a** Correlation matrix of P2's muscle activities. We analyzed the correlation between different muscles based on the data set collected with the rapid stimulation-acquisition system. There is a strong synchronicity between left TA and left GM activities. However, on the right side, they appear to be more independent. **b** Innervation maps of TA for P2. Correlation coefficients of muscle actions and fiber recruitment of different nerve roots were calculated for each participant. Right TA shows a strong correlation with recruitment of L4 and L5 nerve roots while activation of left TA is more correlated with more caudal roots, which is consistent with the results of bipolar stimulation. **c** Innervation maps of GM for P2. The innervation of P2's GM derived from correlation analysis is generally symmetrical on two sides. **d** Bipolar stimulation results indicated the asymmetrical innervation of TA for P2. The configuration using contacts in the 4th row could activate left TA while more caudal configurations were needed to activate right TA.

dataset. Notably, none of the recommended configurations were part of the original training dataset, and the neural network model remained unmodified during this validation process.

Discussion

Here we established the mapping from stimulation to muscle reactions in EES by developing personalized models using a data-driven approach. To better assist patients with SCI restore their motor function, we optimized and fabricated 32-contact neural interfaces based on hybrid simulation. Subsequently, personalized models for two participants were developed based on MRI and CT scans after the implantation of the optimized electrode array. Datasets comprising various stimulation parameters and their corresponding effects were compiled using an automated stimulation-acquisition system. Using networks developed with these data, we achieved reliable quantitative prediction of EES effects on the muscle level for multidimensional stimulation parameters. Furthermore, we employed a dimension-reduction optimization algorithm to recommend parameters based on the model's prediction. Finally, we demonstrate that the parameters recommended by our model could enhance the efficiency of the parameter-tuning process with additional independent datasets while ensuring the feasibility of the algorithmically-driven framework (Fig. 1).

In pursuit of enhanced stimulation selectivity, we refined the design of EES neural interfaces, specifically tailored for the restoration of lower limb motor function. The exact mechanism through which EES facilitates motor function restoration remains elusive, particularly in human subjects[15,16,58–65]. Most neural interfaces used in EES were originally designed for pain suppression[3,5,7,8,20,60,66,67]. However, previous studies indicate the targets of EES for motor restoration may differ from those for pain relief[10,15,16,68–70]. What's more, restoring lower limb motor function with EES often requires coverage of a larger region of the spinal cord[10], necessitating the optimization of neural interfaces. Instead of targeting fibers in the dorsal column[70], large-diameter afferent fibers were indicated to be recruited first in EES for motor function restoration[15,16], which leads to the activation of motor neurons distributed in spinal segments[65]. The spatial selectivity of EES-induced electric fields directly affects muscle activation precision[69]. Neural interfaces with increased contacts and optimized contact arrangement could help improve therapeutic outcomes[10]. However, there is a trade-off between the coverage area and contact density. We conducted computational simulations to optimize the neural interfaces' geometry based on the defined metrics assessing the spatial selectivity of the induced electric field (Fig. 2). We fabricated and implanted 32-contact neural interfaces into two patients with complete motor paralysis and helped them regain the motor function

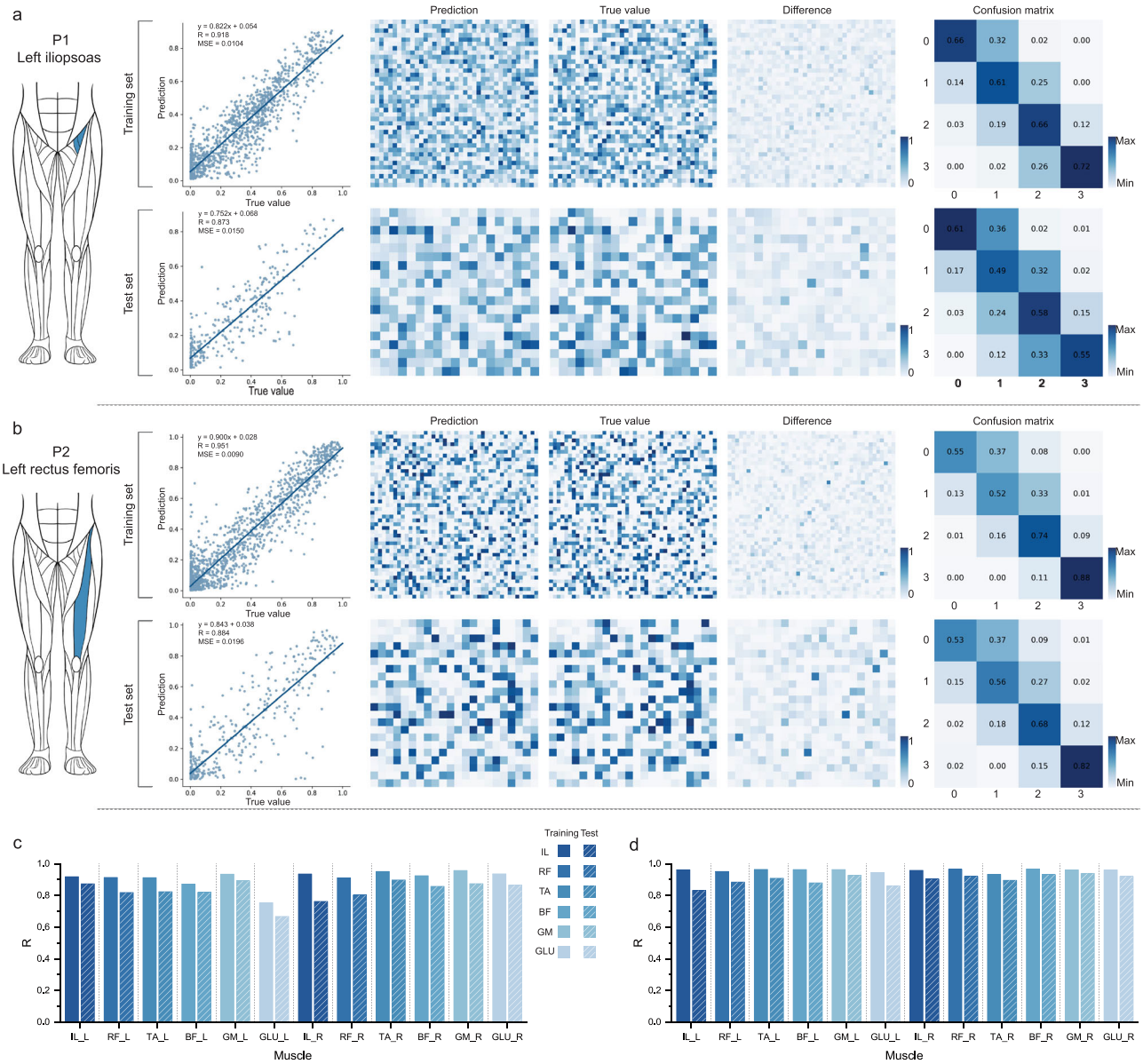


Fig. 5 | Prediction of EES-induced muscle activities. a EMG prediction of the left iliopsoas of P1. The scatter plot indicates the prediction has a strong correlation with true values. Each pixel of matrices in the middle columns represents a sampling and its color reflects the amplitude of evoked EMG. The upper row displays the results of the training set, while the lower row presents the results of the test set. There are obvious similarities between the patterns of prediction and true values. The confusion matrices on the right depict the performance of the model to classify stimulation parameters leading to different response levels. 0 represents almost no muscle response while 3 means the muscle is strongly activated by the stimulation parameter. **b** EMG prediction of the left rectus femoris of P2. Prediction of other channels could be found in supplementary materials. **c**, **d** Pearson correlation coefficients between prediction and true values of two participants. There is a strong correlation between the prediction and true values (For P1, average R among different muscles on test set is 0.829; For P2, average R among different muscle on test set is 0.900).

of their lower limbs successfully. However, a comparative evaluation of the electrode proposed in this study against the existing 32-contact electrode[18] and newly designed electrodes[10] using more appropriate assessment metrics and clinical experiments remains an important but yet-to-be-conducted task. It should also be noted that the altered mechanical properties and decreased system reliability resulting from a higher number of contacts must be taken into consideration. Further studies are needed to determine the optimal contact density that balances between improving spatial resolution and maintaining mechanical flexibility and long-term reliability in vivo.

Our simulation results demonstrated that increasing the contact density can improve stimulation selectivity under unipolar configurations. This finding is consistent with previous reports showing that denser electrode arrays achieve higher spatial selectivity[71,72]. Although the present study focused on unipolar stimulation, it is reasonable to expect that multipolar configurations may further benefit from higher contact density, as suggested by prior modeling studies[10,73]. Nevertheless, substantial inter-individual variability in spinal cord anatomy[10] indicates that a uniform electrode layout may not be optimal for all subjects. Future studies incorporating patient-specific spinal geometries could help refine electrode spacing and alignment to achieve improved selectivity across individuals.

Establishing the complete innervation mapping between structure and function is crucial to enhancing the efficacy of neuromodulation therapies. Prior anatomical knowledge is used to narrow the huge selection space[8]. However, the inter-individual differences always demand manual fine-tuning for each participant. Another widely adopted method to establish spatial mapping of EES involves intraoperative and postoperative testing

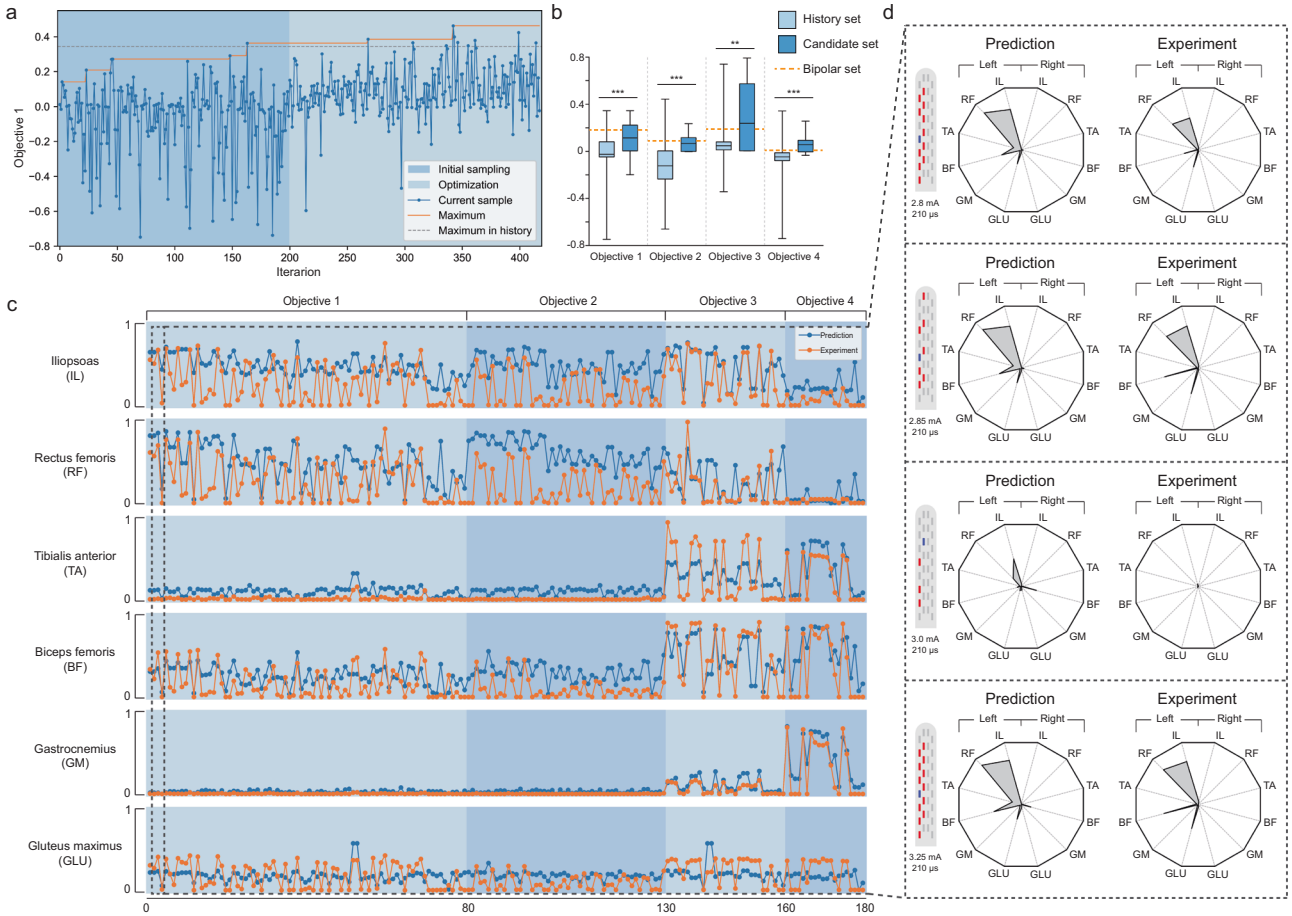


Fig. 6 | Model-based optimization and validation of stimulation parameters. **a** Optimization of Objective 1 based on models' prediction. Two hundred parameters were randomly sampled to initialize the optimization process. The algorithm recommended 20 parameters per batch and the model would predict corresponding muscle actions. Parameters were filtered to form a candidate parameter set based on their predicted performance. **b** Clinical performance of candidate sets. Candidate parameters were delivered to P1. Whiskers of the box-plot represent the maximum and minimum of the achieved objective values. The boxes contain the data from 25% to 75%, and the lines in the box display the mean of the dataset. Orange dash lines show the optimal objectives achieved by bipolar configurations. Candidate sets with less parameters could achieve higher maximums on objective 1 and objective 3 and higher means on all 4 objectives compared with the history set. $^{***}P < 0.001$, $^{**}P < 0.01$. Two-tail student's t-test (objective 1: $P = 2.15 \times 10^{-15}$; objective 2: $P = 1.65 \times 10^{-26}$; objective 3: $P = 1.11 \times 10^{-3}$; objective 4: $P = 9.56 \times 10^{-5}$). **c** Clinical validation of candidate parameters. EMG induced by candidate parameters were recorded and compared with model's prediction. **d** Recorded EMG and corresponding prediction of first four candidate parameters.

and observation[21,26]. Researchers have also tried to predict induced muscle reactions based on preoperative functional MRI and precise reconstruction of patients' spinal cords[10]. A transformation matrix was used to derive motor pool recruitment from calculated dorsal root recruitment, which was developed either based on the averaged location of motor pools across the human population[23], or the projectome of proprioceptive neurons identified from functional MRI. The techniques were employed to design new electrode arrays and identify the optimal implantation position[10]. However, these existing methods may fail to capture more detailed biological characteristics and establish the quantitative mappings for different individuals, especially in the face of neural interfaces with more contacts. In this work, we advance to predict the effects of stimulation at the muscle level by employing a data-driven approach, which was explored to predict stimulation outcomes in other scenarios[74–79]. We used a neural network rather than linear operators[8,10] to capture the intricate neurobiological processes occurring between nerve fiber recruitment and muscle reactions. A multilayer perceptron, inspired by biological neural networks, could effectively address complex input-output relationships by adjusting its synaptic weights and activation functions. Activation matrices calculated from accurate modeling and simulation were used to represent the evoked neural responses, which integrate information of polarity combinations, amplitudes, and pulse widths. Clinical validation suggests that this biologically interpretable dimension-reduction representation proves more effective than end-to-end prediction[25,80]. The hybrid computational model could successfully predict muscle activation to a wide range of stimulation parameters accurately, offering a more personalized and adaptive strategy to deal with the challenge of stimulation-anatomy-function alignment.

Another choice was to directly predict muscle activation from stimulation parameters. The problem faced with this strategy is that it is not easy to find a representation of the discrete polarity combination settings. One choice is to represent the polarity combination as a 32-dimensional vector, using 1 to indicate the anodal contacts, -1 to indicate the cathodal contacts, and 0 to indicate that the contact is not selected. The disadvantage of this representation is that it cannot effectively characterize the positional information of the contacts. For example, if two contacts, one near the cathode and one far from the cathode, are chosen as anodes, then more current tends to flow through the nearer contact than the farther one. The current distribution cannot be represented in the 32-dimensional vector. By modeling and simulation, we used the activation matrix to extract spatial information from the electric field distribution and also integrate information about the stimulus amplitude and pulse width.

We employed the personalized model as a reliable agent to search for feasible stimulation parameters based on its prediction. While the method of gradually increasing stimulation amplitudes during testing effectively

mitigates the risk from excessive stimulation, it can also impede the efficiency of the parameter-tuning process. After the definition of optimization objectives for targeted movements, the personalized model can recommend stimulation parameters within the vast parameter space employing algorithms. This minimizes the need for sequential testing and provides clinicians with reliable quantitative references and insights. The developed automated acquisition system efficiently gathered multi-channel EMG data while ensuring controlled risks. In our experiments, we could collect 12-channel EMG for over 1500 configurations within 1 hour and no discomfort from participants was reported during the data acquisition process (Fig.S11).

One limitation of this study is the variability observed in EMG responses induced by the same stimulation parameters, which brings challenges to the robustness of models. The slight decrease in prediction accuracy on additional datasets and bipolar datasets confirms this hypothesis (Figs. S13–16, 23). While EMG was chosen to characterize muscle reactions due to its convenience and patient-friendly nature, fluctuations of approximately 10% were observed even when delivering the same stimulation within a short time interval. The underlying factors contributing to this variability remain unclear and may involve measurement inaccuracies or physiological factors such as muscle fatigue and transient neuromuscular instability. Averaging EMG responses across multiple stimulation pulses to obtain more reliable representations of muscle activation is necessary in future studies. Additionally, changes in posture significantly influence stimulation effects[81–85]. We observed that a greater angle between the torso and the horizontal direction resulted in weaker responses, possibly due to alterations in the distance between the spinal cord and the electrode array (Fig. S24). Over a longer time scale, the drift of the implanted electrode array and tissue growth around it will also affect stimulation effects for the same parameter. More accurate and stable acquisition of stimulation effects need to be considered. Additionally, the gap between EMG amplitudes and body movement can't be ignored, either. Developing the mapping between muscle activation and dynamic performance to define more practical objectives became the next task. We used the recruitment ratio of multi-diameter axons as the representation of neural activation induced by EES. As the understanding of human EES progresses, it is also necessary to propose more comprehensive representations that encompass the responses of both motor neurons and interneurons[16,63]. Moreover, there is still room for improvement in the accuracy of the modeling. Our current model includes only the tissues within the spinal canal and does not account for structures such as vertebrae and fibers outside the cerebrospinal fluid (CSF). Although some studies indicate that these extradural tissues have a limited impact on simulation accuracy[39–42], incorporating them into the model is important for surgical planning of implant procedures. Our simulations indicate that axonal recruitment under EES is sensitive to the modeled axonal trajectories, as extending proximal trajectories leads to systematically altered recruitment patterns and lower activation thresholds under identical stimulation settings (Fig. S8). Additional quantitative analyses further showed that a substantial fraction of simulated fibers exhibited action potential initiation at proximal terminal nodes, although this proportion decreased when the axonal trajectories were extended (Fig. S9). This effect arises from axonal truncation and imposed boundary conditions, which cause action potentials to initiate at proximal terminal nodes and subsequently propagate along the axon[86]. Such end-nodal recruitment represents a numerical artifact and does not fully reflect physiological activation mechanisms[10,15]. Accordingly, conclusions derived from these simulation results may be affected by model-dependent biases and do not fully capture the complexity of physiological axonal activation. Although subsequent data-driven components of the framework may partially compensate for this systematic bias, these findings underscore the necessity of more complete and anatomically accurate modeling of neural circuits to improve the physiological interpretability of simulation-based predictions.

This study is based on the assumption that EES activates motor neurons primarily by first stimulating afferent nerve fibers, thereby leading to motor neuron activation or increased excitability[8,10,15,16]. Consequently, we used the recruitment ratio of nerve roots as a primary input to the neural network. However, alternative hypotheses regarding the mechanism of EES exist, such as its potential role in modulating the state of spinal circuitry[3,4,6,7]. Developing more comprehensive and detailed neural network models may contribute to a deeper understanding of the underlying mechanisms of EES.

In the future, the proposed parameter tuning method requires more extensive clinical validation and comparison with expert-guided approaches. Integrating this method with emerging technologies (e.g., pre-operative fMRI[10] and post-operative fMRI with MRI conditional systems[87]) to incorporate more prior knowledge into the model and constrain the parameter search space could further enhance its robustness and efficiency.

We derived innervation alignment between nerve roots and muscles based on experimental measurements and simulation results (Fig. 4, S17, 18). Correlation analysis provides more quantitative understanding for spinal innervation and muscle synergy in EES. In Fig. 4, fibers with diameter of 7.3 μm and 5.7 μm appear to play a dominant role in muscle activation, which contradicts existing findings that large-diameter fibers are primarily responsible for muscle activation[8,15]. One possible reason for this inconsistency is that, in our simulations, we deliberately increased the stimulation intensity by a fixed ratio to avoid cases where few fibers would be activated due to low stimulation intensity, leading to the activation of small-diameter fibers. This approach allowed the activation matrix to more comprehensively reflect both the spatial distribution and intensity of stimulation. However, the accuracy of this method still requires further validation via more direct clinical experiments. Improving simulation and algorithmic efficiency to enable real-time prediction and visualization of stimulation outcomes is another advancement which could be considered in the future.

In conclusion, our study shows that the muscle activities evoked by various EES parameters could be accurately predicted. Personalized innervation mappings between stimulation and muscle activation were established in a data-driven manner, which could help develop custom treatment strategies. The hybrid model could be seen as a digital twin of the patient, reducing the need of extensive on-site parameter testing. By tailoring diverse optimization objectives and carrying out large-scale model-based parameter exploration, patient-specific parameter libraries could be created. Additionally, once a comprehensive mapping between stimulation and muscle responses is established, we can assess the patient's muscle synergy and selectivity, providing valuable guidance for personalized training and treatment. The challenge of over-expansion of the parameter space, which arises with an increase in complexity of neural interfaces, can be addressed with the assistance of artificial intelligence, enabling more precise and natural neuromodulation. The AI-aided framework could be applied in other neuromodulation therapies such as deep brain stimulation (DBS), vagus nerve stimulation (VNS), and transcranial magnetic stimulation (TMS). By establishing a predictive mapping between stimulation parameters and physiological responses via simulation and neural network modeling, this framework can assist in parameter optimization, electrode design, and closed-loop control—paving the way toward personalized and intelligent neuromodulation strategies.

Data availability

The source data for Fig. 2 is in Supplementary Data 1 and the source data for Fig. 6 could be found in Supplementary Data 2. The detailed data for two participants can be found in supplementary materials and GitHub (https://github.com/LHD-LHD/Predict-EES).

Code availability

The code for parameter search and recommendation could be found online (https://github.com/yunyuewei/HdSafeBO). Upon reasonable request, the data and code used in this study will be available to investigators from the corresponding author using private online cloud storage for reproducibility analyses.

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References

1. Carhart, M. R., He, J. P., Herman, R., D'Luzansky, S. & Willis, W. T. Epidural spinal-cord stimulation facilitates recovery of functional walking following incomplete spinal-cord injury. *Ieee T Neur. Sys. Reh.* **12**, 32–42 (2004).
2. Minassian, K. et al. Stepping-like movements in humans with complete spinal cord injury induced by epidural stimulation of the lumbar cord: electromyographic study of compound muscle action potentials. *Spinal Cord.* **42**, 401–416 (2004).
3. Harkema, S. et al. Effect of epidural stimulation of the lumbosacral spinal cord on voluntary movement, standing, and assisted stepping after motor complete paraplegia: a case study. *Lancet* **377**, 1938–1947 (2011).
4. Angeli, C. A., Edgerton, V. R., Gerasimenko, Y. P. & Harkema, S. J. Altering spinal cord excitability enables voluntary movements after chronic complete paralysis in humans. *Brain* **137**, 1394–1409 (2014).
5. Danner, S. M. et al. Human spinal locomotor control is based on flexibly organized burst generators. *Brain* **138**, 577–588 (2015).
6. Grahn, P. J. et al. Enabling task-specific volitional motor functions via spinal cord neuromodulation in a human with paraplegia. *Mayo Clin. Proc.* **92**, 544–554 (2017).
7. Gill, M. L. et al. Neuromodulation of lumbosacral spinal networks enables independent stepping after complete paraplegia (vol 24, pg 1677, 2018). *Nat. Med.* **24**, 1942–1942 (2018).
8. Wagner, F. B. et al. Targeted neurotechnology restores walking in humans with spinal cord injury. *Nature* **563**, 65-+ (2018).
9. Kandhari, S. et al. Epidural spinal stimulation enables global sensorimotor and autonomic function recovery after complete paralysis: 1 study from India. *IEEE Trans. Neural Syst. Rehab. Eng.* **30**, 2052–2059 (2022).
10. Rowald, A. et al. Activity-dependent spinal cord neuromodulation rapidly restores trunk and leg motor functions after complete paralysis. *Nat. Med.* **28**, 260 (2022).
11. Lorach, H. et al. Walking naturally after spinal cord injury using a brain-spine interface. *Nature* **618**, 126 (2023).
12. Powell, M. P. et al. Epidural stimulation of the cervical spinal cord for post-stroke upper-limb paresis. *Nat. Med.* **29**, 689 (2023).
13. Milekovic, T. et al. A spinal cord neuroprosthesis for locomotor deficits due to Parkinson's disease. *Nat. Med.* **29**, 2854 (2023).
14. Squair, J. W. et al. Implanted system for orthostatic hypotension in multiple-system atrophy. *N. Engl. J. Med.* **386**, 1339–1344 (2022).
15. Capogrosso, M. et al. A computational model for epidural electrical stimulation of spinal sensorimotor circuits. *J. Neurosci.* **33**, 19326–19340 (2013).
16. Moraud, E. M. et al. Mechanisms underlying the neuromodulation of spinal circuits for correcting gait and balance deficits after spinal cord injury. *Neuron* **89**, 814–828 (2016).
17. Won, S. M., Song, E. M., Reeder, J. T. & Rogers, J. A. Emerging modalities and implantable technologies for neuromodulation. *Cell* **181**, 115–135 (2020).
18. Romeni S., et al. High-frequency epidural electrical stimulation reduces spasticity and facilitates walking recovery in patients with spinal cord injury. *Sci. Transl. Med.* **17**, 9607 (2025).
19. Parker S. R., et al An active electronic bidirectional interface for high resolution interrogation of the spinal cord. *bioRxiv*, 2024.2005.2029.596250 (2024).
20. Angeli, C. A. et al. Recovery of over-ground walking after chronic motor complete spinal cord injury. *N. Engl. J. Med.* **379**, 1244–1250 (2018).
21. Angeli, C. et al. Targeted selection of stimulation parameters for restoration of motor and autonomic function in individuals with spinal cord injury. *Neuromodul. Technol. Neural Interface* **27**, 645–660 (2023).
22. Sharrard W., The segmental innervation of the lower limb muscle in man. *Ann. R. Coll. Surg. Engl.* **35**, 106–122 (1964).
23. Schirmer, C. M. et al. Heuristic map of myotomal innervation in humans using direct intraoperative nerve root stimulation. *J. Neurosurg.-Spine* **15**, 64–70 (2011).
24. Phillips, L. H. & Park, T. S. Electrophysiologic mapping of the segmental anatomy of the muscles of the lower-extremity. *Muscle Nerve* **14**, 1213–1218 (1991).
25. Kachuee M., et al An active learning based prediction of epidural stimulation outcome in spinal cord injury patients using dynamic sample weighting. *IEEE International Conference on Healthcare Informatics* (*ICHI*), 478-483 (IEEE, 2017).
26. Hoglund, B. K. et al. Mapping spinal cord stimulation-evoked muscle responses in patients with chronic spinal cord injury. *Neuromodulation* **26**, 1371–1380 (2023).
27. Zhang, X. et al. Voluntary walking related joint movement training with targeted epidural electrical stimulation enabled neural recovery for individuals with spinal cord injury. *Sci. Bull.* **69**, 3507–3511 (2024).
28. Thomson, A. Fifth annual report of the committee of collective investigation of the Anatomical Society of Great Britain and Ireland for the year 1893-94. *J. Anat. Physiol.* **29**, 35 (1894).
29. McCotter, R. E. Regarding the length and extent of the human medulla spinalis. *Anat. Rec.* **10**, 559–564 (1916).
30. Kameyama, T., Hashizume, Y. & Sobue, G. Morphologic features of the normal human cadaveric spinal cord. *Spine* **21**, 1285–1290 (1996).
31. Zhou, M. W. et al. Microsurgical Anatomy of Lumbosacral Nerve Rootlets for Highly Selective Rhizotomy in Chronic Spinal Cord Injury. *Anat. Rec.-Adv. Integr. Anat. Evolut. Biol.* **293**, 2123–2128 (2010).
32. Yujing, D. & Tianzhong, Z. Study of proportion spinal cord segment in Chinese. *J. Lanzhou Univ. Med. Sci.* **2**, 16–21 (1985).
33. Frostell A., Hakim R., Thelin E. P., Mattsson P., Svensson M. A review of the segmental diameter of the healthy human spinal cord. *Front. Neurol* **7**, 238 (2016).
34. Boonpirak, N. & Apinhasmit, W. Length and caudal level of termination of the spinal-cord in Thai adults. *Acta Anat.* **149**, 74–78 (1994).
35. Bai H., Chen W., Dai D., Zhang M., The sagittal and transverse diameters of Chinese spinal canal. *Acta Anatomica Sinica* (WHO, 1955).
36. Mendez, A. et al. Segment-specific orientation of the dorsal and ventral roots for precise therapeutic targeting of human spinal cord. *Mayo Clin. Proc.* **96**, 1426–1437 (2021).
37. Elvan, Ö, Aktekin, M. & Kayan, G. Microsurgical anatomy of the spinal cord in human fetuses. *Surg. Radio. Anat.* **42**, 951–960 (2020).
38. Zannou, A. L. et al. Temperature increases by kilohertz frequency spinal cord stimulation. *Brain Stimul.* **12**, 62–72 (2019).
39. Zander H. J., Graham R. D., Anaya C. J., Lempka S. F. Anatomical and technical factors affecting the neural response to epidural spinal cord stimulation. *J. Neural Eng.* **17**, 036019 (2020).
40. Holsheimer, J. Computer modelling of spinal cord stimulation and its contribution to therapeutic efficacy. *Spinal Cord.* **36**, 531–540 (1998).
41. Hernández-Labrado, G. R., Polo, J. L., López-Dolado, E. & Collazos-Castro, J. E. Spinal cord direct current stimulation: finite element analysis of the electric field and current density. *Med. Biol. Eng. Comput.* **49**, 417–429 (2011).
42. Troughton J. G., Ansong Snr Y. O., Duobaite N., Proctor C. M. Finite element analysis of electric field distribution during direct current stimulation of the spinal cord: implications for device design. *APL Bioeng.* **7**, 046109 (2023).
43. McIntyre, C. C. & Grill, W. M. Extracellular stimulation of central neurons: Influence of stimulus waveform and frequency on neuronal output. *J. Neurophysiol.* **88**, 1592–1604 (2002).
44. Schiefer, M. A., Triolo, R. J. & Tyler, D. J. A model of selective activation of the femoral nerve with a flat interface nerve electrode for a lower extremity neuroprosthesis. *IEEE Trans. Neural Syst. Rehab.Eng.* **16**, 195–204 (2008).
45. Raspopovic, S., Capogrosso, M. & Micera, S. A computational model for the stimulation of rat sciatic nerve using a transverse intrafascicular

multichannel electrode. *IEEE Trans. Neural Syst. Rehab.Eng.* **19**, 333–344 (2011).
46. Ladenbauer, J., Minassian, K., Hofstoetter, U. S., Dimitrijevic, M. R. & Rattay, F. Stimulation of the human lumbar spinal cord with implanted and surface electrodes: a computer simulation study. *IEEE Trans. Neural Syst. Rehab.Eng.* **18**, 637–645 (2010).
47. Rattay, F. Analysis of models for external stimulation of axons. *IEEE Trans. Bio-Med. Eng.* **33**, 974–977 (1986).
48. McIntyre, C. C., Richardson, A. G. & Grill, W. M. Modeling the excitability of mammalian nerve fibers: Influence of afterpotentials on the recovery cycle. *J. Neurophysiol.* **87**, 995–1006 (2002).
49. Richardson, A. G., McIntyre, C. C. & Grill, W. M. Modelling the effects of electric fields on nerve fibres: influence of the myelin sheath. *Med Biol. Eng. Comput* **38**, 438–446 (2000).
50. Hines, M. L., Morse, T., Migliore, M., Carnevale, N. T. & Shepherd, G. M. ModelDB: a database to support computational neuroscience. *J. Comput Neurosci.* **17**, 7–11 (2004).
51. Lloyd, S. P. Least-squares quantization in PCM. *IEEE Trans. Inf. Theory* **28**, 129–137 (1982).
52. Hines, M. L. & Carnevale, N. T. The NEURON simulation environment. *Neural Comput* **9**, 1179–1209 (1997).
53. Dali M., et al. Model based optimal multipolar stimulation without knowledge of nerve structure: application to vagus nerve stimulation. *J. Neural Eng.* **15**, 046018 (2018).
54. Cogan, S. F. Neural stimulation and recording electrodes. *Annu. Rev. Biomed. Eng.* **10**, 275–309 (2008).
55. Merrill, D. R., Bikson, M. & Jefferys, J. G. R. Electrical stimulation of excitable tissue: design of efficacious and safe protocols. *J. Neurosci. Meth* **141**, 171–198 (2005).
56. Y. Wei, Z. Yi, H. Li, S. Soedarmadji, Y. Sui, paper presented at the Proceedings of The 8th Conference on Robot Learning, (PMLR, 2025).
57. Yao, Q. et al. Redistribution of intraspinal and muscular 18F-FDG uptake after the epidural electrical stimulation treatment in spinal cord injury individuals. *Clin. Nuclear Med.* **50**, 401–406 (2025).
58. Hachmann, J. T. et al. Epidural spinal cord stimulation as an intervention for motor recovery after motor complete spinal cord injury. *J. Neurophysiol.* **126**, 1843–1859 (2021).
59. Eisdorfer J. T., et al. Epidural electrical stimulation: a review of plasticity mechanisms that are hypothesized to underlie enhanced recovery from spinal cord injury with stimulation. *Front. Mol. Neurosci.* **13**, 163 (2020).
60. Formento, E. et al. Electrical spinal cord stimulation must preserve proprioception to enable locomotion in humans with spinal cord injury. *Nat. Neurosci.* **21**, 1728 (2018).
61. Pino I. P., et al. Long-term spinal cord stimulation after chronic complete spinal cord injury enables volitional movement in the absence of stimulation. *Front. Syst. Neurosci.* **14**, 35 (2020).
62. Beck, L. et al. Impact of long-term epidural electrical stimulation enabled task-specific training on secondary conditions of chronic paraplegia in two humans. *J. Spinal Cord. Med.* **44**, 800–805 (2021).
63. Kathe, C. et al. The neurons that restore walking after paralysis. *Nature* **611**, 540-+ (2022).
64. Squair, J. W. et al. Recovery of walking after paralysis by regenerating characterized neurons to their natural target region. *Science* **381**, 1338–1345 (2023).
65. Minassian, K., Hofstoetter, U., Tansey, K. & Mayr, W. Neuromodulation of lower limb motor control in restorative neurology. *Clin. Neurol. Neurosurg.* **114**, 489–497 (2012).
66. Herman, R., He, J., D'Luzansky, S., Willis, W. & Dilli, S. Spinal cord stimulation facilitates functional walking in a chronic, incomplete spinal cord injured. *Spinal Cord.* **40**, 65–68 (2002).
67. Barolat, G., Myklebust, J. B. & Wenninger, W. Enhancement of Voluntary Motor Function Following Spinal-Cord Stimulation - Case-Study. *Appl Neurophysiol.* **49**, 307–314 (1986).
68. Wenger N., et al. Closed-loop neuromodulation of spinal sensorimotor circuits controls refined locomotion after complete spinal cord injury. *Sci. Transl. Med.* **6**, 133 (2014).
69. Wenger, N. et al. Spatiotemporal neuromodulation therapies engaging muscle synergies improve motor control after spinal cord injury. *Nat. Med.* **22**, 138–145 (2016).
70. Molnar, G. & Barolat, G. Principles of cord activation during spinal cord stimulation. *Neuromodulation* **17**, 12–21 (2014).
71. Jantz M. K., et al. High-density spinal cord stimulation selectively activates lower urinary tract nerves. *J. Neural Eng.* **19**, 066014 (2022).
72. Gad P., et al. Development of a multi-electrode array for spinal cord epidural stimulation to facilitate stepping and standing after a complete spinal cord injury in adult rats. *J. Neuroeng. Rehabil.* **10**, 2 (2013).
73. Cuellar C., et al. Selective activation of the spinal cord with epidural electrical stimulation. *Brain Sci* **14**, 650 (2024).
74. Golabek J., SchiefervM., Wong J. K., Saxena S., Patrick E., Artificial neural network-based rapid predictor of biological nerve fiber activation for DBS applications. *J. Neural Eng.* **20**, 16 (2023).
75. Romeni S., et al. Combining biophysical models and machine learning to optimize implant geometry and stimulation protocol for intraneural electrodes. *J. Neural Eng.* **20**, 219 (2023).
76. Hussain M. A., Grill W. M., Pelot N. A. Highly efficient modeling and optimization of neural fiber responses to electrical stimulation. *Nat. Commun.* **15**, 7597 (2024).
77. Zhao, Z. X. et al. Optimization of spinal cord stimulation using Bayesian preference learning and its validation. *IEEE Trans. Neural Syst. Rehab.* **29**, 1987–1997 (2021).
78. Phokaewvarangkul O., Vateekul P., Wichakam I., Anan C., Bhidayasiri R. Using machine learning for predicting the best outcomes with electrical muscle stimulation for tremors in Parkinson's disease. *Front. Aging Neurosci.* **13**, 727654 (2021).
79. Boutet A., et al. Predicting optimal deep brain stimulation parameters for Parkinson's disease using functional MRI and machine learning. *Nat. Commun.* **12**, 3043 (2021).
80. Feldman E. R., Burdick J. W., Modeling motor responses of paraplegics under epidural spinal cord stimulation. *Proc. IEEE Embs C Neur E*, 354-357 (IEEE2017).
81. Kuechmann, C., Valine, T. & Wolfe, D. L. Could automatic position adaptive stimulation be useful in spinal cord stimulation. (Blackwell Publishing Ltd., 2009).
82. Cameron, T. & Alo, K. M. Effects of posture on stimulation parameters in spinal cord stimulation. *Neuromodulation* **1**, 177–183 (1998).
83. North, R. B., Sung, J. H., Matthews, L. A., Zander, H. J. & Lempka, S. F. Postural changes in spinal cord stimulation thresholds: current and voltage sources. *Neuromodulation* **27**, 178–182 (2024).
84. Schade, C. M., Schultz, D., Tamayo, N., Iyer, S. & Panken, E. Automatic adaptation of neurostimulation therapy in response to changes in patient position: results of the posture responsive spinal cord stimulation (PRS) research study. *Pain. Physician* **14**, 407–417 (2011).
85. Parker, J. L., Karantonis, D. M., Single, P. S., Obradovic, M. & Cousins, M. J. Compound action potentials recorded in the human spinal cord during neurostimulation for pain relief. *Pain* **153**, 593–601 (2012).
86. Reilly, J. P. Survey of numerical electrostimulation models. *Phys. Med Biol.* **61**, 4346–4363 (2016).
87. Birthi P., et al. Advancements in MRI conditionality of spinal cord stimulation systems: a narrative review of recent SCS systems and their associated risks in MRI operations. *Pain Phys.* **28**, 451–465 (2025).

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Competing interests

The authors declare no competing interests.

Additional information

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